Introduction to Robotics — Homework 1

AR-601M arm: kinematics, differential kinematics and singularities

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1 Robot and scope

AR-601M is an anthropomorphic robot with 41 degrees of freedom. This assignment concerns one arm only: a serial chain with six actuated joints from the shoulder to the tool. The geometry used in the assignment is shown in Figures 1 and 2. The joint variables are

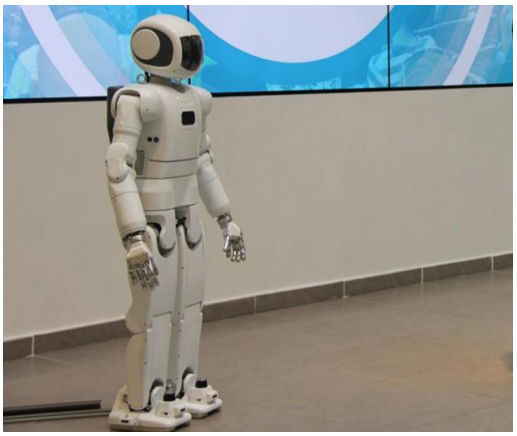


Figure 1: AR-601M robot. The assignment covers one six-joint arm.

$$q = [q_1, q_2, q_3, q_4, q_5, q_6]^T.$$

The required results are a kinematic model, forward and inverse kinematics, differential kinematics, and a numerical map of singular configurations. The choice of coordinate frames, the mathematical representation of the chain, and the solution methods are part of the assignment. They must be defined and justified in the report.

2 Mechanism data and constraints

2.1 Joint order and axes

The joints are revolute and occur in the following order from the base to the tool. Each axis is expressed in the local frame associated with that joint.

Joint	Axis	Lower limit	Upper limit
q_1	z	-90°	$+90^\circ$
q_2	y	-105°	$+105^\circ$
q_3	x	-45°	$+45^\circ$
q_4	z	-90°	$+90^\circ$
q_5	y	-90°	$+90^\circ$
q_6	x	-90°	$+90^\circ$

The limits above are closed intervals. A configuration outside any interval is infeasible.

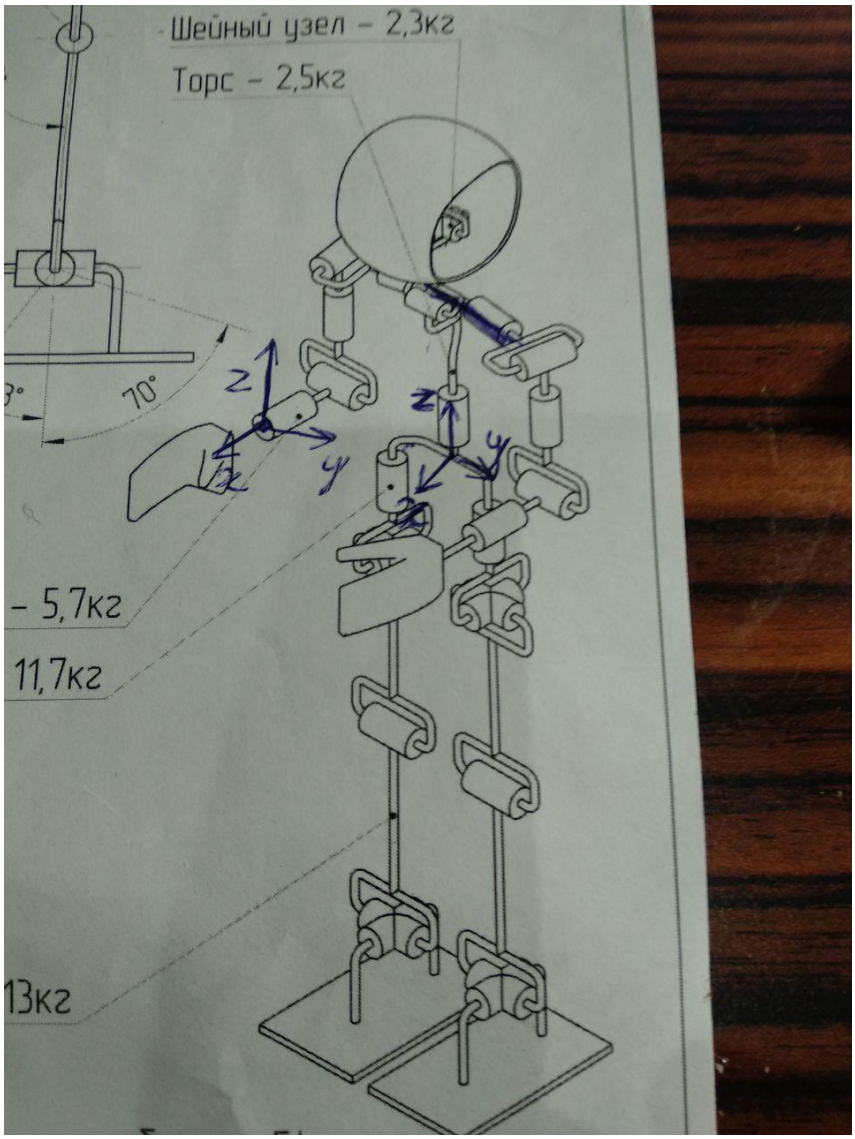


Figure 2: Arm geometry and axis annotations used to construct the kinematic model.

2.2 Fixed geometry

The chain contains four fixed offsets:

Symbol	Offset	Value
l_1	base to shoulder, along $+z$	0.40 m
l_2	shoulder offset, along $-by$	0.18 m
l_3	arm offset, along $-z$	0.30 m
l_4	wrist centre to tool, along $+x$	0.12 m

Here $b = +1$ denotes the right arm and $b = -1$ denotes the left arm. The left arm is the mirror image of the right arm in the x–z plane. Therefore the sign of the l_2 offset changes with b. A fixed rotation of 10° about z is located between joints q_1 and q_2. Its sign is $+10^\circ$ for $b = +1$ and -10° for $b = -1$. This rotation is a fixed geometric parameter, not an additional joint. Unless a task states otherwise, all translations are expressed in the local frame at their position in the serial chain. Angles used in program code must be converted consistently to radians.

2.3 Common input data

All students use the same geometric parameters and joint limits throughout Tasks 1–6. No student-specific parameters or alternative link dimensions are used.

3 Required software interface

The submitted implementation must provide these operations or direct language-equivalent interfaces:

- `FK(q, robot) -> T`, where $T \in SE(3)$;
- `IK(T, robot) -> list[q]`;
- `DifferentialKinematics(q, robot) -> J`, where $J \in \mathbb{R}^{6\times 6}$ maps joint velocities to tool linear and angular velocities.

The argument `robot` must be loaded from a separate, human-readable model file. The model file must contain the joint order, joint axes, joint limits, fixed rotation, link offsets, arm side b, and link lengths. Values must not be duplicated as hidden constants in the solver code.

4 Tasks

Task 1. Kinematic model Construct the coordinate-frame model of the arm and encode it in the robot model file. Required evidence:

- a diagram of the frames, joint axes and fixed offsets;
- the complete symbolic transformation chain from base to tool;
- the model-file contents or a report excerpt containing all model parameters;
- a program check that loads exactly six revolute joints in the stated order and with the stated limits.

Task 2. Forward kinematics Implement FK. Evaluate it for:

- the zero configuration;
- one configuration with at least one joint at a limit;
- one generic configuration strictly inside all joint limits.

Required evidence for every configuration:

- the joint vector q and resulting matrix T;
- $\|R^T R - I\|$ and $|\det R - 1|$;
- the tolerance used for both checks.

Additionally report the generic pose orientation as ZYX roll–pitch–yaw angles. Use the same three configurations in all subsequent checks where applicable. Provide a visualizer: enter six joint angles and show the corresponding manipulator drawing and numerical homogeneous matrix T. Include an image and matrix for each of the three configurations.

Task 3. Analytical inverse kinematics Derive and implement an analytical inverse-kinematics solver. The solver must return all distinct real solution branches before joint-limit filtering and all feasible branches after filtering. Required evidence:

- the derivation used in the implementation;
- results for both $b = +1$ and $b = -1$;
- explicit handling of unreachable and degenerate targets;
- the complete branch list for each test target;
- for each returned branch, the residual between the requested target and FK(q, robot);
- a stated rule for identifying duplicate branches.

At least one test target must be generated from a generic feasible joint vector so that reachability is known. The visualizer must also accept that FK pose or a target position and ZYX Euler

angles ($R = R_z(\psi)R_y(\theta)R_x(\phi)$), then show every distinct feasible manipulator configuration and print every corresponding six-angle combination. Show degenerate or continuous families separately from a finite list.

Task 4. Numerical inverse kinematics Implement a numerical inverse-kinematics solver and apply it to the same targets used in Task 3. Use at least two initial configurations that converge to different solution branches. Required evidence for every run:

- initial configuration;
- termination status;
- number of iterations;
- final configuration;
- final position and orientation residuals;
- nearest analytical branch from Task 3.

Failed runs must remain in the reported results and must include the failure condition.

Task 5. Differential kinematics Derive and implement the differential kinematics of the arm. The result must be a 6×6 matrix $J(q)$ satisfying

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q},$$

where v and ω are the tool linear and angular velocities expressed in the frame stated in the report. Obtain $J(q)$ by two independent derivations and compare both implementations with a finite-difference approximation based on FK. Required evidence:

- the two derivations;
- the finite-difference definition and step size;
- the configuration used for comparison;
- pairwise matrix residuals for all three results;
- one additional comparison at a configuration near a singularity.

Task 6. Singularity analysis Sample the feasible joint space and evaluate a stated singularity metric derived from J at every sample. Produce a map or set of plots sufficient to identify singular regions. Required evidence:

- the sampling ranges, resolution and total number of samples;
- the definition and numerical threshold of the singularity metric;
- the resulting plot or plots with labelled axes and units;
- at least one reported singular or near-singular configuration;
- a stick-figure drawing of that configuration showing all joint axes and fixed offsets;
- verification of the reported configuration using the differential kinematics matrix J.

The report must state that a joint-space sampling may map several different configurations to the same Cartesian point; consequently, a Cartesian plot of such samples can contain overlapping branches and is not a one-to-one representation of the configuration space. Add a Cartesian workspace slice with a continuous color scale counting distinct feasible IK configurations from zero to the maximum attained for the stated case. State the fixed orientation, arm side, joint limits, slice and numerical tolerance. Show an unreachable point as zero; mark unresolved cases and continuous families separately from the finite color scale.

5 Submission format

Submit one repository containing all of the following:

- `pixi.toml` or `Dockerfile` at the repository root;

- the robot model file;
- source code implementing Tasks 1–6 and the required visualizations;
- automated tests;
- a PDF or Markdown report covering Tasks 1–6 in order;
- a README containing the single command required to reproduce all reported numerical results and figures from a clean checkout.

The submission must run with one of these forms:

- `pixi run check`; or
- documented `docker build` and `docker run` commands.

The report must contain the following information for every task:

- input data and arm side;
- assumptions and conventions;
- requested derivation, numerical result or figure;
- verification result and tolerance;
- a reference to the source file or test that reproduces the result.

Results that cannot be reproduced from the submitted repository are not accepted. Missing branches, omitted failed runs, unstated tolerances, or figures without defined axes and sampling parameters are treated as missing task requirements.

6 Pre-submission checklist

- The repository runs from a clean checkout using the documented command.
- The model file contains six joints and all fixed geometric parameters.
- The report states the arm side used for each result.
- Every reported IK branch is checked by forward kinematics.
- Both differential-kinematics implementations and the finite-difference result are compared.
- All plots contain labelled axes, units, legends where needed, and sampling data.
- Every numerical value and figure in the report is generated by submitted code.